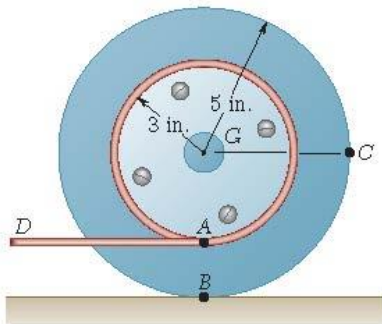
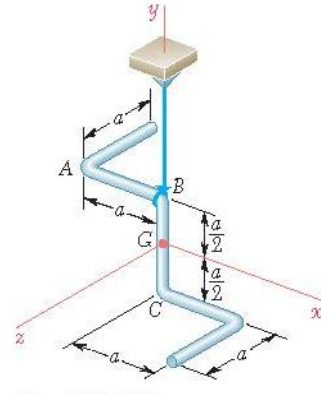


Basics. Check the following statement correct or not, and explain why.

- S1. A uniform rod of mass m and length $5a$ is bent into the shape (see figure right). It rotates about y -axis with the angular rotation rate, ω , the kinetic energy of the body is $T = \frac{1}{2}I_y\omega^2$ where I_y is the moment of the inertia of the body about the axis, y .

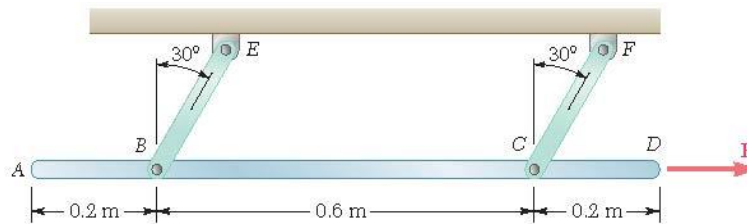


- S2. A 3-in.-radius drum is rigidly attached to a 5-in. radius drum as shown left. One of the drums rolls without sliding on the surface shown, and a cord is wound around the other drum. The friction force from ground on the wheel is $\mu_k N$ where μ_k is the kinetic friction coefficient between the drum and surface, and N is the normal supporting force from ground surface to the drum.

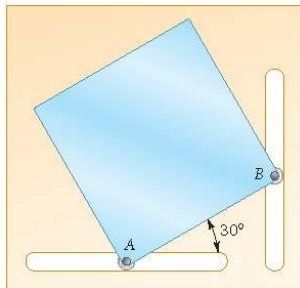
- S3. Two balls on a smooth horizontal surface collide to each other, the total kinetic energy of the two-balls system after collision will be reduced.

Problem 1. At the instant shown the angular velocity of links BE and CF is 6 rad/s counterclockwise and is decreasing at the rate of 12 rad/s². Knowing that the length of each link is 300 mm and neglecting the weight of the links, and the mass of rod AD is 6 kg.

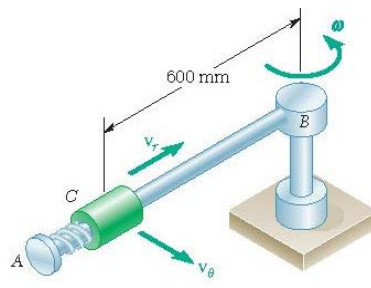
- Determine the velocities of B and D, and type of the motion for the rod ABCD,
- draw the free-body and motion diagrams of the rod ABCD,
- determine the force P, and
- the corresponding forces in links BE and CF.



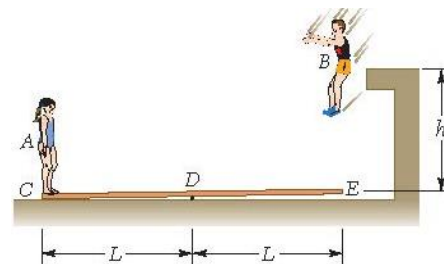
Problem 2. The motion of a square plate of side 150 mm and mass 2.5 kg is guided by pins at corners A and B that slide in slots cut in a vertical wall. Immediately after the plate is released from rest in the position shown, determine (a) the angular acceleration of the plate, and (b) the reaction at corner A. The force-motion diagram is required. Hint: initially the angular velocities of A and B are zero.



Problem 2



Problem 3



Problem 4

Problem 3. Collar C has a mass of 8 kg and can slide freely on rod AB, which in turn can rotate freely in a horizontal plane. The assembly is rotating with an angular velocity ω of 1.5 rad/s when a spring located between A and C is released, projecting the collar along the rod with an initial relative speed $v_r = 1.5$ m/s. Knowing that the combined mass moment of inertia about B of the rod and spring is 1.2 kg m², determine (a) the minimum distance between the collar and point B in the ensuing motion, (b) the corresponding angular velocity of the assembly. Noted: It is preferred to use the principle of work and energy to solve this problem.

Problem 4. The plank CDE has a mass of 15 kg and rests on a small pivot at D. The 55-kg gymnast A is standing on the plank at C when the 70-kg gymnast B jumps from a height of 2.5 m and strikes the plank at E. Assuming perfectly plastic impact and that gymnast A is standing absolutely straight, determine (a) the velocity of gymnast B right before striking the plank, (b) the velocity of gymnast A right after gymnast B striking the plank, and (c) the height to which gymnast A will rise.